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ALKHIDMAT FOUNDATION PAKISTAN  
PROJECT REQUIREMENT DOCUMENT

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# Emergency Blood Response Platform

**Branded as: QATRA**

*Connecting Verified Seekers to Eligible Donors — in Minutes, Not Hours*

|                        |                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------|
| <b>PROJECT NAME</b>    | Emergency Blood Response System (branded as QATRA)                                      |
| <b>ORGANIZATION</b>    | Alkhidmat Foundation Pakistan                                                           |
| <b>GROUP NUMBER</b>    | 01                                                                                      |
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| <b>DOCUMENT STATUS</b> | Approved by Team                                                                        |

## Team Workload & Section Assignments

| Team Member                                              | Assigned Sections | Primary Feature / Scope                                 |
|----------------------------------------------------------|-------------------|---------------------------------------------------------|
| Abdul Hayy Khan                                          | Section 1, 2, & 8 | Overall System Architecture, Goals, & Final Assembly    |
| Abdul Hayy Khan<br>(Completed on behalf of Hareem Israr) | Section 3         | Feature 1: Live Map Integration & Proximity Matching    |
| Saghir Ahmed                                             | Section 4         | Feature 2: Authentication, Authorization & Verification |
| Mahrukh Baig                                             | Section 5         | Feature 3: Social Media Feed for Blood Donation         |
| Yumna Abbasi                                             | Section 6         | Feature 4: Awareness Sessions & Eligibility Module      |
| Nimra Iftikhar                                           | Section 7         | Non-Functional Requirements, Security & KPIs            |

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# Section 1: Project Overview & Goals

## 1.1 Executive Summary

Every year, families across Karachi and other major Pakistani cities face the same crisis: a relative is admitted for emergency surgery, a trauma case, a delivery complication, or a chronic condition like thalassemia, and the hospital tells them blood is needed now. The default response is a WhatsApp broadcast — forwarded across group chats, often losing context, arriving too late, or reaching people who cannot verify whether the request is even real. Alkhidmat Foundation Pakistan sees this pattern constantly through its community and campus networks, and the Emergency Blood Response Platform is being built to replace that broadcast-and-hope model with something structured, verified, and fast.

The platform connects three groups that currently operate through disconnected, informal channels: emergency blood seekers (patients' families and hospital staff), verified donors, and campus or community drive organizers who run the pre-screened donor pipelines Alkhidmat depends on. At its core is a hospital-slip-and-CNIC verification layer that keeps commercial resellers and fake requests off the network, a live proximity-matching map that finds eligible donors within a configurable radius, a structured social feed that replaces WhatsApp spam with filterable, actionable posts, and an awareness/eligibility module that keeps the donor pool educated and pre-screened before they ever show up to a drive.

Version 1.0 is scoped around Karachi trauma centers and Alkhidmat's existing campus chapters, with the explicit goal of proving the core loop — verified request in, matched and eligible donor out, in under 15 minutes — before expanding geographically or adding logistics like blood bag delivery.

## 1.2 Core Problems Addressed

- Emergency search bottlenecks: WhatsApp broadcast chains cause hours of delay, message loss, and duplicated or contradictory information at exactly the moment speed matters most.
- Donor trust & verification gaps: seekers cannot tell a legitimate hospital request from a scam or a commercial blood reseller, and donors have no reliable way to confirm a request is real before showing up.
- Social & cultural awareness barriers: myths about donor eligibility, gender-related taboos, and general unfamiliarity with the donation process shrink the usable donor pool and cause repeated on-site disqualifications at drives.

## 1.3 Project Goals (In-Scope)

- Goal 1: Reduce emergency search-to-connection time to under 15 minutes for urgent requests.
- Goal 2: Ensure 100% of live requests are verified via hospital admission slips and CNIC validation before broadcast.
- Goal 3: Establish a pre-screened, cooldown-tracked donor pool that eliminates avoidable on-site drive disqualifications.
- Goal 4: Give community and campus drive leads a centralized tool to run and staff their own blood drives instead of manual WhatsApp coordination.

## **1.4 Non-Goals (Out-of-Scope for Version 1.0)**

- No physical blood bag delivery, courier, or cold-chain logistics — the platform matches people and requests, it does not move blood.
- No paid or commercial blood transactions of any kind; the platform is strictly a free, non-commercial matching service.
- No direct integration with hospital electronic medical record (EMR) systems in V1 — verification relies on slip upload + OCR + admin review, not live hospital data feeds.
- No geographic expansion beyond Karachi trauma centers and existing Alkhidmat campus chapters in this version.

## Section 2: User Personas & Core Journeys

### 2.1 Unified Persona: Emergency Seeker & Donor

| Attribute              | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Profession & Age Group | Working professional / family decision-maker, aged 25–45                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Life Context           | Active on local WhatsApp networks; frequently handles household emergencies and community welfare requests on behalf of extended family                                                                                                                                                                                                                                                                                                                                                                                                          |
| Core Frustration       | <p><i>“Emergency ke waqt WhatsApp broadcast bhej ke ghanthon wait karna padta hai... logon ke replies nahi aate, fake requests aur privacy leaks ka darr hota hai, aur hospital jao to donors eligibility issue ya pehle se arrangement ki wajah se reject ho jate hain.”</i></p> <p>In short: broadcasting on WhatsApp during an emergency means hours of waiting with no replies, real fear of fake requests and privacy leaks, and donors who do turn up getting turned away at the hospital for eligibility issues or prior commitments.</p> |
| Key Needs              | Instant proximity matching within a 10–15 km radius, authenticated hospital requests, contact privacy, and automated pre-screening so a donor who shows up is actually eligible.                                                                                                                                                                                                                                                                                                                                                                 |

### 2.2 Secondary Persona: Society / Campus Drive Lead (Proxy)

| Attribute              | Detail                                                                                                                                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Profession & Age Group | Student leader / NGO campus chapter president, aged 21–22                                                                                                                      |
| Core Frustration       | Operational energy wasted manually verifying outdated donor contacts and handling turn-aways on drive day due to age/CNIC issues or recent health disqualifications.           |
| Key Needs              | A centralized, pre-verified database of eligible donors with automated 90-day cooldown tracking and event management tools, so drive day doesn't turn into on-the-spot triage. |

### 2.3 Core User Journeys

#### Journey A — Emergency Seeker

1. Family member is told at the hospital that a patient urgently needs blood.
2. Seeker registers/logs in via Phone OTP (CNIC already on file or captured at this step).
3. Seeker photographs and uploads the hospital admission/requisition slip.



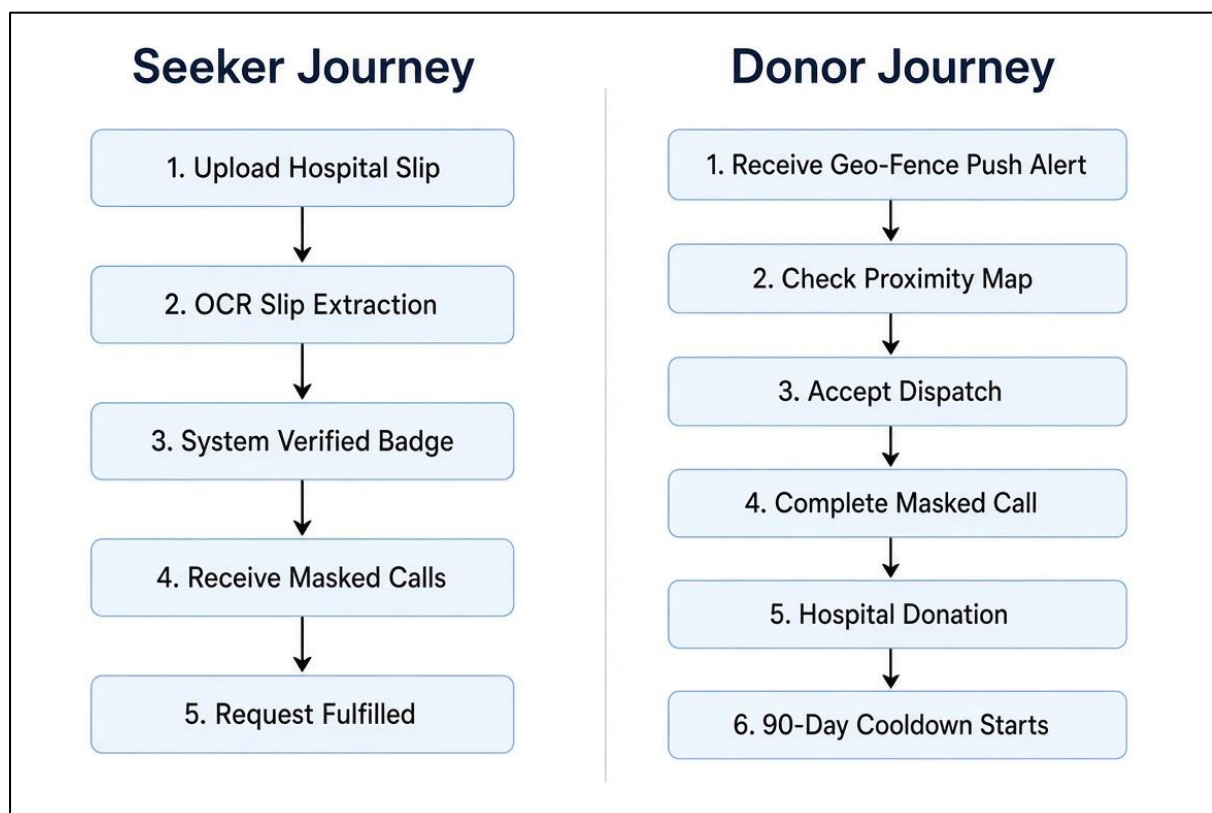
4. System OCR-validates the slip; on success the request goes live on the map and feed within minutes.
5. Eligible, nearby donors are alerted; seeker receives responses through the masked in-app channel, not their real number.
6. Request auto-closes once required units are confirmed fulfilled.

### **Journey B — Verified Donor**

1. Donor completes CNIC + OTP verification and the pre-screening health checklist once, at signup.
2. Donor receives a geo-fenced push alert for a verified request matching their blood group within their radius.
3. Donor reviews the request card (hospital, urgency, distance) and accepts.
4. Donor is connected to the seeker via masked call/chat and travels to the hospital.
5. Post-donation, the donor's 90-day cooldown timer starts automatically and they are excluded from further alerts until it resets.

### **Journey C — Campus Drive Lead**

1. Drive lead schedules a campus blood drive through the Awareness & Event module.
2. Interested students self-register as donor or volunteer and, where relevant, run the eligibility checker before signing up.
3. Drive lead sees a pre-verified, pre-screened attendee list ahead of the event, cutting down manual phone-call verification.
4. On drive day, the lead scans attendee QR codes instead of manually re-checking CNIC/age at the door.



## Section 3: Feature 1 — Live Map Integration & Proximity Matching

### 3.1 Feature Overview

The live map is the real-time backbone connecting verified emergency requests to eligible donors. Every verified request (post-Section-4 approval) is plotted at its hospital's coordinates; every donor who opts into live matching is geo-indexed so the system can instantly compute who is close enough, and eligible enough, to be alerted. This replaces the guesswork of a WhatsApp broadcast — where nobody knows who actually saw the message or how far away they are — with a ranked, distance-sorted, eligibility-filtered dispatch list the system generates automatically.

The map serves two audiences differently: seekers see a status view of their own request (donors alerted, donors responding, distance/ETA once a donor accepts), while donors see an alert-driven view — they are not expected to browse the map hunting for requests; the system pushes matching requests to them and the map is where they confirm distance and route before committing.

### 3.2 Functional Requirements

#### FR 1.1 — Donor Location Tracking

- FR 1.1.1: System captures a donor's live or last-known location only while “Available to Donate” is toggled on; location sharing is off by default at signup.
- FR 1.1.2: Location updates are sent on a throttled interval (e.g. every 2–5 minutes or on significant movement) rather than continuous streaming, to conserve battery and data.
- FR 1.1.3: Donors can pause or fully revoke location sharing at any time from a single visible toggle; pausing immediately removes them from active matching pools.

#### FR 1.2 — Request Map Display

- FR 1.2.1: Every verified request is plotted at its hospital's geocoded coordinates (not the seeker's personal location, to avoid exposing a private address).
- FR 1.2.2: Markers are color-coded by urgency (e.g. red = within 2 hours, orange = within 24 hours, gray = fulfilled/closed) so the map is scannable at a glance.
- FR 1.2.3: Tapping a marker opens a summary card — blood group, units needed, urgency, distance from the viewer — without exposing seeker identity or hospital-internal details beyond what Section 4 already approved for public display.

#### FR 1.3 — Geo-Fenced Alert Dispatching

- FR 1.3.1: On request approval, the system queries donors within a default 10 km radius who (a) match the required blood group or are compatible per standard donor-recipient compatibility rules, (b) are not in an active 90-day cooldown, and (c) have passed pre-screening (Section 4.3).
- FR 1.3.2: Radius is configurable per request at creation (5 km / 10 km / 15 km) to account for hospital density — dense urban hospitals may default tighter, peripheral ones wider.
- FR 1.3.3: Auto-expansion — if fewer than a minimum threshold of eligible donors (e.g. 5) are found within the initial radius after a set wait window (e.g. 5 minutes with no

acceptance), the system automatically expands the radius by a fixed increment (e.g. +5 km) and re-dispatches, up to one or two expansion cycles.

- FR 1.3.4: Rare blood groups bypass the wait window and expand immediately, consistent with the rare-group escalation logic in Section 5.3.

#### **FR 1.4 — Proximity-Ranked Matching Output**

- FR 1.4.1: The donor list returned for a request is sorted by ascending distance, computed via the haversine formula against last-known donor coordinates.
- FR 1.4.2: The ranking additionally deprioritizes (but does not exclude) donors who have already declined this specific request, so a decline doesn't silently blackhole the donor from future unrelated requests.
- FR 1.4.3: Once a donor accepts, all other pending alerts for that request are marked as “being fulfilled” on their end, so multiple donors don't independently rush to a request that's about to close.

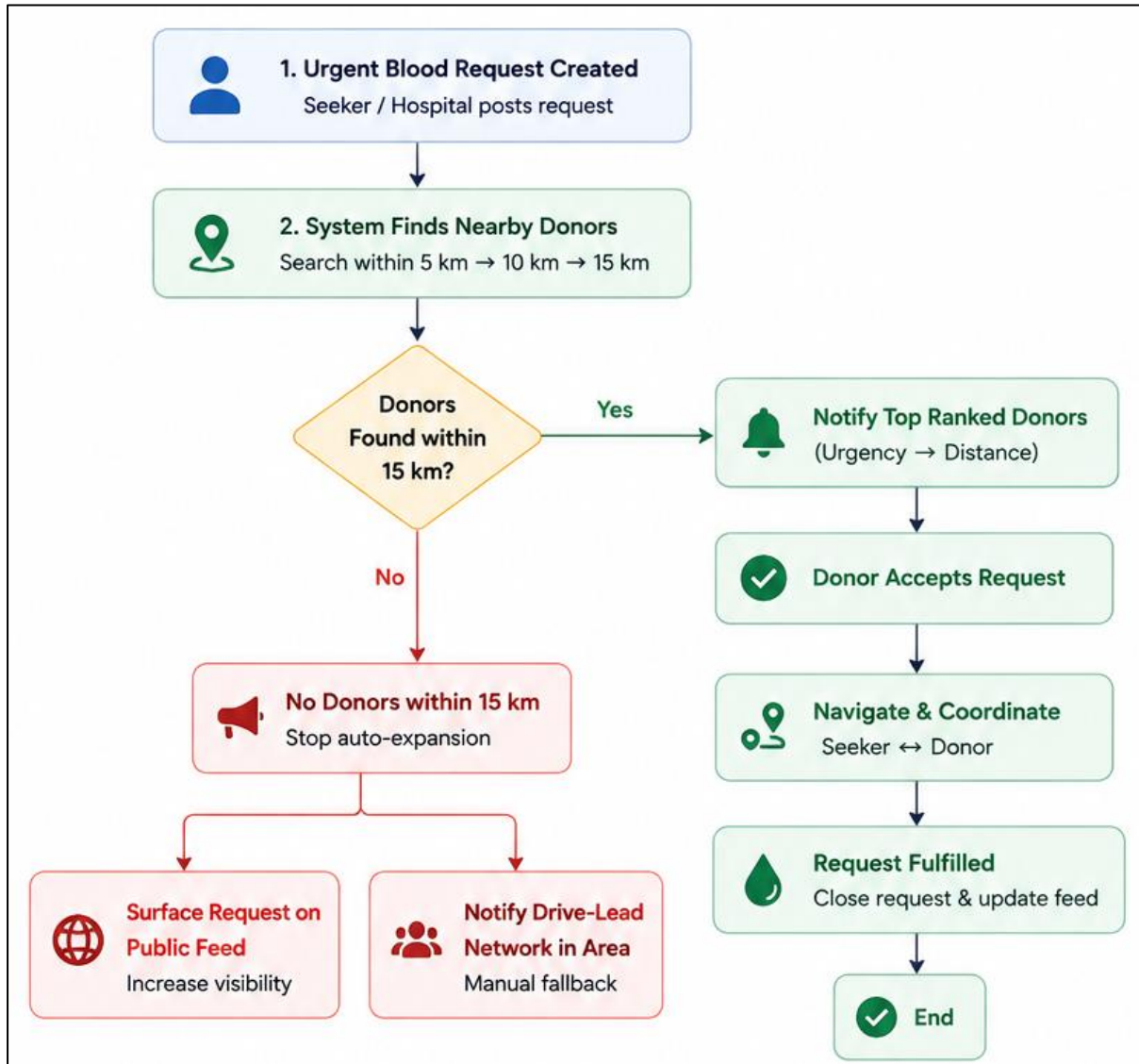
### **3.3 Technical & API Requirements**

- Map rendering & geocoding: Google Maps Platform (Maps SDK + Geocoding API) as primary, with OpenStreetMap/Leaflet evaluated as a lower-cost fallback if API cost at 10k-concurrent scale becomes a constraint.
- Geo-indexed backend store: PostGIS (Postgres extension) or a geohash-based indexing scheme, shared with the infrastructure described in NFR 1.3 (Section 7) — nearest-donor lookups must stay fast as the donor pool grows into the thousands, not degrade into a full-table scan.
- Distance calculation: haversine formula for proximity sorting; radius queries executed as bounded geo-index range queries against the geo-index, not application-side filtering of a full donor list.
- Push delivery: alert dispatch integrates with the platform's push notification service (shared with Section 5.3's notification triggers) — the map module computes \*who\* to alert, notification delivery infrastructure is shared, not duplicated.
- Location data handling: donor coordinates are treated as sensitive and are not persisted longer than needed for active matching (see NFR 2.2, Section 7, on keeping sensitive data isolated) — historical location trails are not retained.

### **3.4 User Flow & UI Edge Cases**

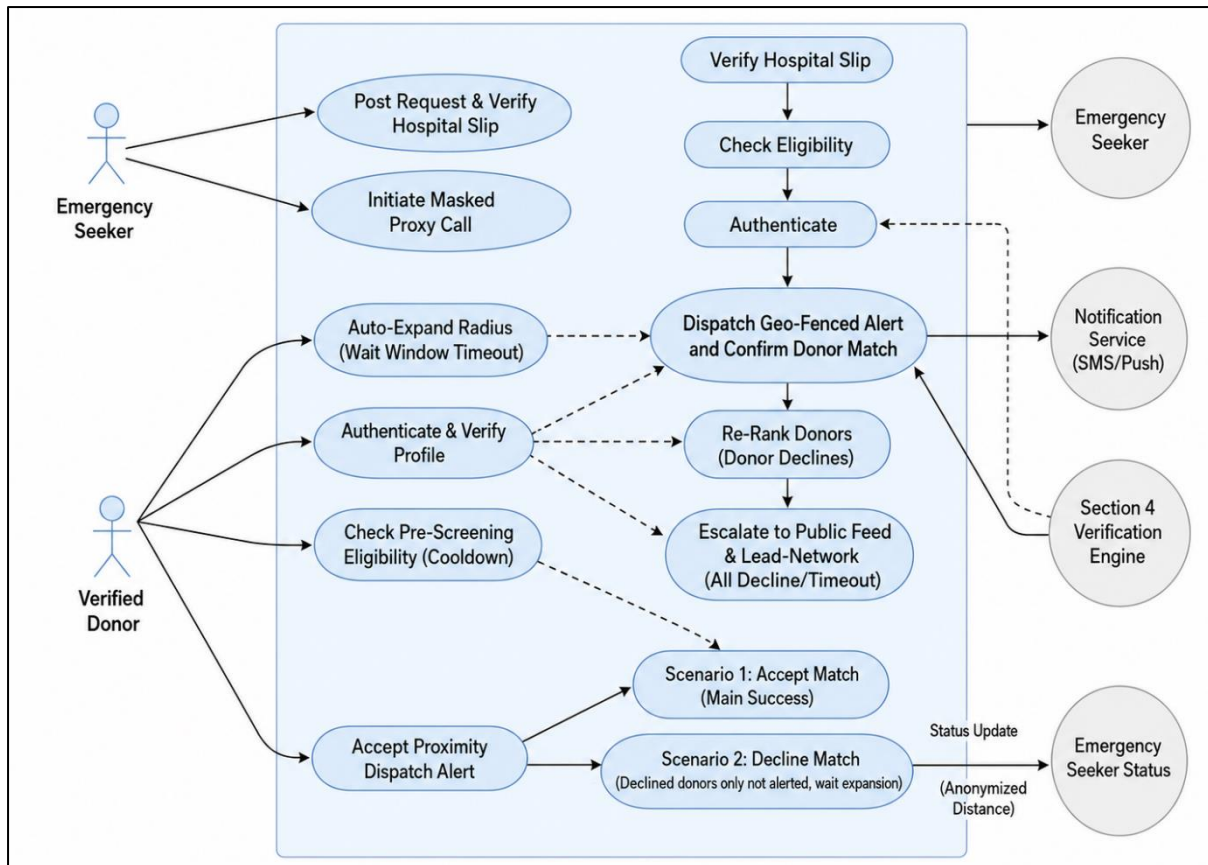
- No donors found within 15 km after all expansion cycles: system stops auto-expanding, surfaces the request more prominently on the public feed (Section 5), and notifies the drive-lead network in that area as a manual fallback.
- Donor disables location permissions: donor can still browse the feed and manually opt in to specific requests, but is excluded from automatic geo-fenced push alerts until location sharing is re-enabled.
- Donor location is stale beyond a threshold (e.g. 30 minutes since last update): system falls back to last-known location for ranking purposes but flags it as stale, deprioritizing that donor slightly versus donors with a fresh location fix.

- Donor accepts but then can't actually make it (traffic, sudden unavailability): donor can cancel acceptance from the map view, which immediately re-opens the request to the next-ranked donor rather than leaving the seeker waiting on a silent no-show.
- Multiple simultaneous urgent requests near the same donor: donor sees a stacked alert list ranked by urgency-then-distance, not just the most recent push.



### 3.5 Core Use Case: Donor Receives and Responds to a Proximity Alert

- Use Case ID: UC-MAP-01
- Title: Dispatch Geo-Fenced Alert and Confirm Donor Match
- Primary Actor: Verified Donor
- Secondary Actors: Emergency Seeker, Notification Service, Section 4 Verification Engine
- Preconditions: request has passed Section 4 verification; donor has location sharing and “Available to Donate” enabled.
- Postconditions: donor is matched to the request and proxy contact is established (handoff to Section 4.7's proxy call bridging), or the donor declines and the next-ranked donor is alerted.



### Main Success Scenario

1. System receives a newly verified request and computes eligible donors within the configured radius.
2. Eligible donors are ranked by distance and filtered against cooldown/pre-screening status.
3. Top-ranked donors receive a push alert showing blood group, urgency, hospital, and distance.
4. Donor opens the alert and views the request on the map, including estimated distance/route.
5. Donor accepts; system marks the request as “matched,” notifies other alerted donors that it's being fulfilled, and hands off to the masked proxy-contact flow.
6. Seeker sees the request status update from “Searching” to “Donor Matched” with anonymized distance/ETA.

### Alternative & Exception Flows

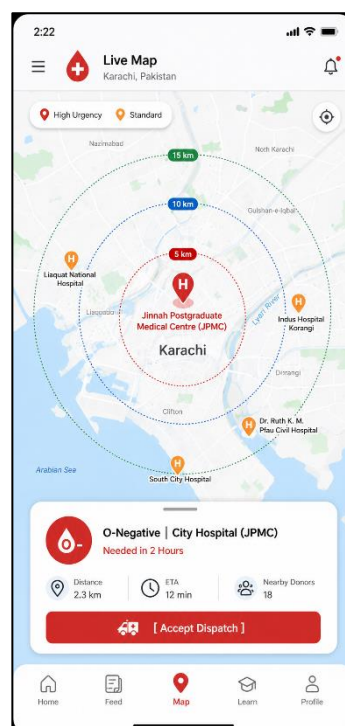
- Alternative Flow A (No response within wait window): system triggers the radius auto-expansion described in FR 1.3.3 and re-dispatches to a wider donor pool.
- Alternative Flow B (Donor declines): request is re-ranked without that donor for this specific request; donor remains eligible for future unrelated alerts.
- Exception Flow C (All nearby donors decline or timeout): request is escalated to the public feed and relevant drive-lead network, per FR 1.4 fallback behavior.

### 3.6 Module KPIs

| Metric                    | Target                                      | Why It Matters                                                                                             |
|---------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Average Match Radius Used | $\leq 10$ km for 80% of fulfilled requests  | Validates that the default radius is well-calibrated and expansion cycles are the exception, not the norm. |
| Alert-to-Acceptance Time  | $< 5$ minutes median                        | Directly feeds the platform-wide Time-to-Match KPI in Section 7.                                           |
| Stale-Location Rate       | $< 10\%$ of active donor pool at alert time | High staleness degrades ranking accuracy and donor ETA reliability.                                        |
| Radius Expansion Rate     | $< 15\%$ of requests require expansion      | A high rate signals the donor pool density assumption is wrong for a given area.                           |

### 3.7 Summary

The Live Map Integration & Proximity Matching module functions as the real-time spatial engine of the Emergency Blood Response Platform. By replacing chaotic broadcast forwards with precision location services, the module plots verified hospital requests alongside color-coded urgency indicators and concentric geo-fence radii (5 km, 10 km, 15 km). It automatically queries eligible, non-cooldown donors within proximity using PostGIS spatial indexing, presenting a distance-ranked summary card featuring real-time ETA, distance metrics, and one-tap dispatch acceptance to achieve an under-15-minute emergency response loop.





## **Section 4: Feature 2 — Authentication, Authorization & Verification**

### **4.1 Feature Overview**

The Authentication, Authorization & Verification module is the trust and security foundation of the platform. In emergency medical scenarios, unverified requests and predatory commercial resellers cause critical delays, emotional trauma, and fatal miscommunication. This module establishes a multi-layered verification pipeline to eliminate fraudulent requests, protect user identity, enforce donor cooldown windows, and manage fine-grained system privileges.

Identity is validated through CNIC verification paired with phone OTP authentication. For seekers, an OCR pipeline with manual admin override verifies hospital admission slips before a request is broadcast. For donors, an automated 90-day cooldown timer is synchronized with a pre-screening health questionnaire to prevent on-site drive disqualifications.

#### **Primary Strategic Objectives**

- Eliminate commercial exploitation by requiring dual-key identity verification (CNIC + hospital slip) that unauthorized brokers cannot easily fake.
- Ensure every active emergency signal on the map and feed originates from a legitimate medical facility.
- Protect user privacy through total phone number masking and anonymous proxy routing during initial negotiation.
- Automate donor readiness with precise 90-day cooldown tracking and pre-screen health checklists.

### **4.2 Functional Requirements & Sub-Modules**

#### **FR 2.1 — User Onboarding, Identity Registration & Phone OTP**

- FR 2.1.1 Mobile Phone Authentication: users register with a valid Pakistani (+92) number; system dispatches a 6-digit OTP via SMS gateway, valid for 180 seconds.
- FR 2.1.2 CNIC Identity Binding: users submit their 13-digit CNIC plus front/back photo capture; the system validates the CNIC checksum and extracts key identity attributes.
- FR 2.1.3 Biometric / Selfie Match (optional, high-trust): a real-time liveness check for elevated donor privileges, preventing impersonation with stolen credentials.

#### **FR 2.2 — Emergency Hospital Admission Slip Verification Engine**

- FR 2.2.1 Hospital Slip Capture: seeker uploads a high-resolution photo/scan of the requisition slip, showing hospital letterhead, doctor stamp, patient MRN, required units, and blood group.
- FR 2.2.2 Automated OCR Data Extraction: extracts patient name, hospital name, ward/room, physician name, required components (PRBC/Platelets/Plasma), and unit quantity.
- FR 2.2.3 Rapid Admin Review & Escalation Queue: if OCR confidence is below 85% or stamps are unclear, the request routes to Alkhidmat's 24/7 Verification Desk for a one-tap approve/reject, target response under 3 minutes.

### FR 2.3 — Role-Based Access Control (RBAC)

| User Role                | Assigned Permissions                                                                                                                     | Verification Required                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Guest / Unverified User  | Browse public awareness content, view educational material, test eligibility checker. Cannot view active requests or donor contacts.     | Basic app download; no credentials.                        |
| Verified Seeker          | Create emergency requests, upload requisition slips, track matching status, send masked messages/calls to matched donors.                | Phone OTP + CNIC + approved requisition slip.              |
| Verified Donor           | Receive targeted proximity alerts, toggle live availability, accept/decline requests, view pre-donation health status.                   | Phone OTP + CNIC + pre-screening questionnaire.            |
| Drive / Event Organizer  | Schedule campus/community drives, publish awareness events, track registered volunteers and pre-screened donors, scan attendee QR codes. | Organization credential verification + Alkhidmat approval. |
| System Admin / Desk Lead | Access 24/7 verification queue, manually approve/reject slips, ban fraudulent accounts, view analytics, manage system flags.             | Multi-factor hardware key / internal corporate SSO.        |

## 4.3 Automated Donor Eligibility & Cooldown Tracking

### FR 2.4 — Automated 90-Day Cooldown Engine

- Automated calculation: on confirmed donation completion, the platform locks donor status and starts a live 90-day countdown.
- Proximity alert exclusion: donors under active cooldown are dynamically excluded from geo-fenced broadcasts.
- Cooldown reset & push triggers: a Day-85 pre-notification, followed by a status update to “Eligible & Active” at Day 90.

### FR 2.5 — Interactive Pre-Screening Health Checklist

| Screening Metric | Standard Eligibility Threshold             | System Action If Failed                                |
|------------------|--------------------------------------------|--------------------------------------------------------|
| Age Criteria     | 18–65 years old                            | Hard locks donor registration profile.                 |
| Minimum Weight   | ≥ 50 kg (110 lbs)                          | Displays health warning & soft-disqualifies donor.     |
| Hemoglobin Level | ≥ 12.5 g/dL (self-reported / field tested) | Recommends dietary consultation; flags for field test. |

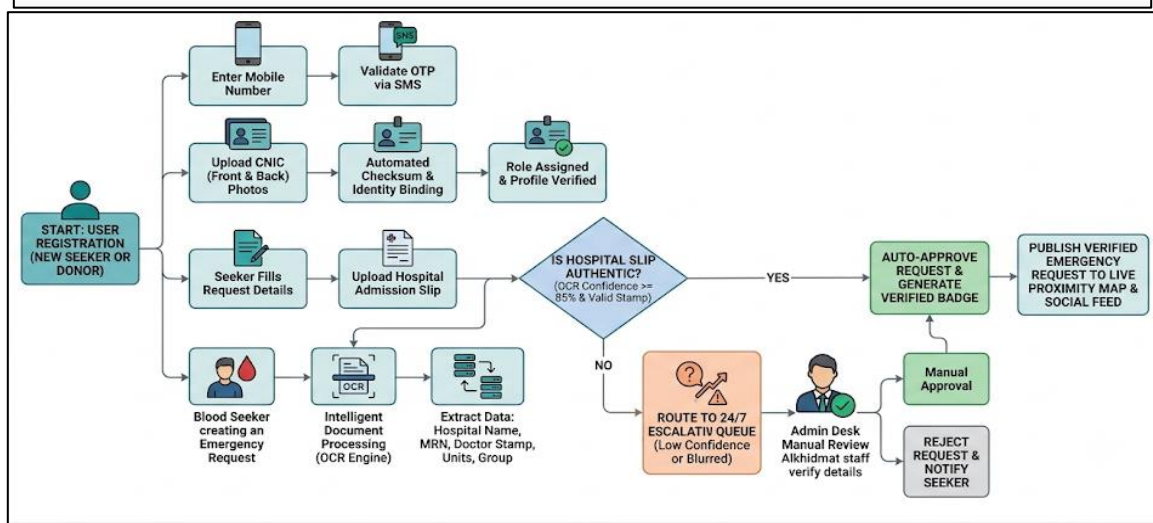
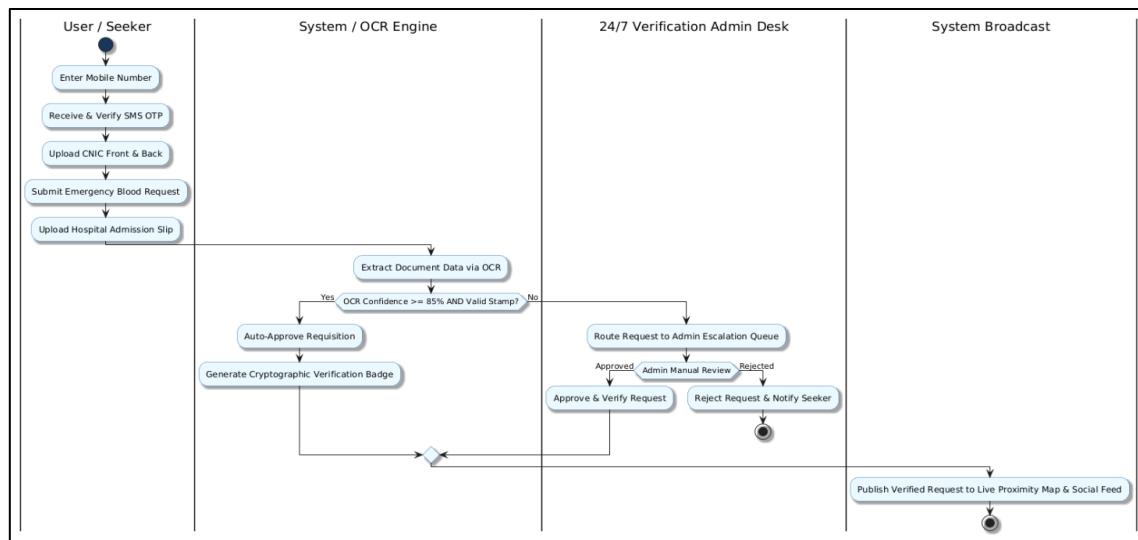


| Screening Metric            | Standard Eligibility Threshold                             | System Action If Failed                             |
|-----------------------------|------------------------------------------------------------|-----------------------------------------------------|
| Recent Illness / Medication | No fever, flu, or antibiotic usage in past 14 days         | Applies temporary 14-day hold on availability.      |
| Travel / Tattoo History     | No major surgical procedures or tattooing in last 6 months | Applies 6-month deferral status with auto-reminder. |

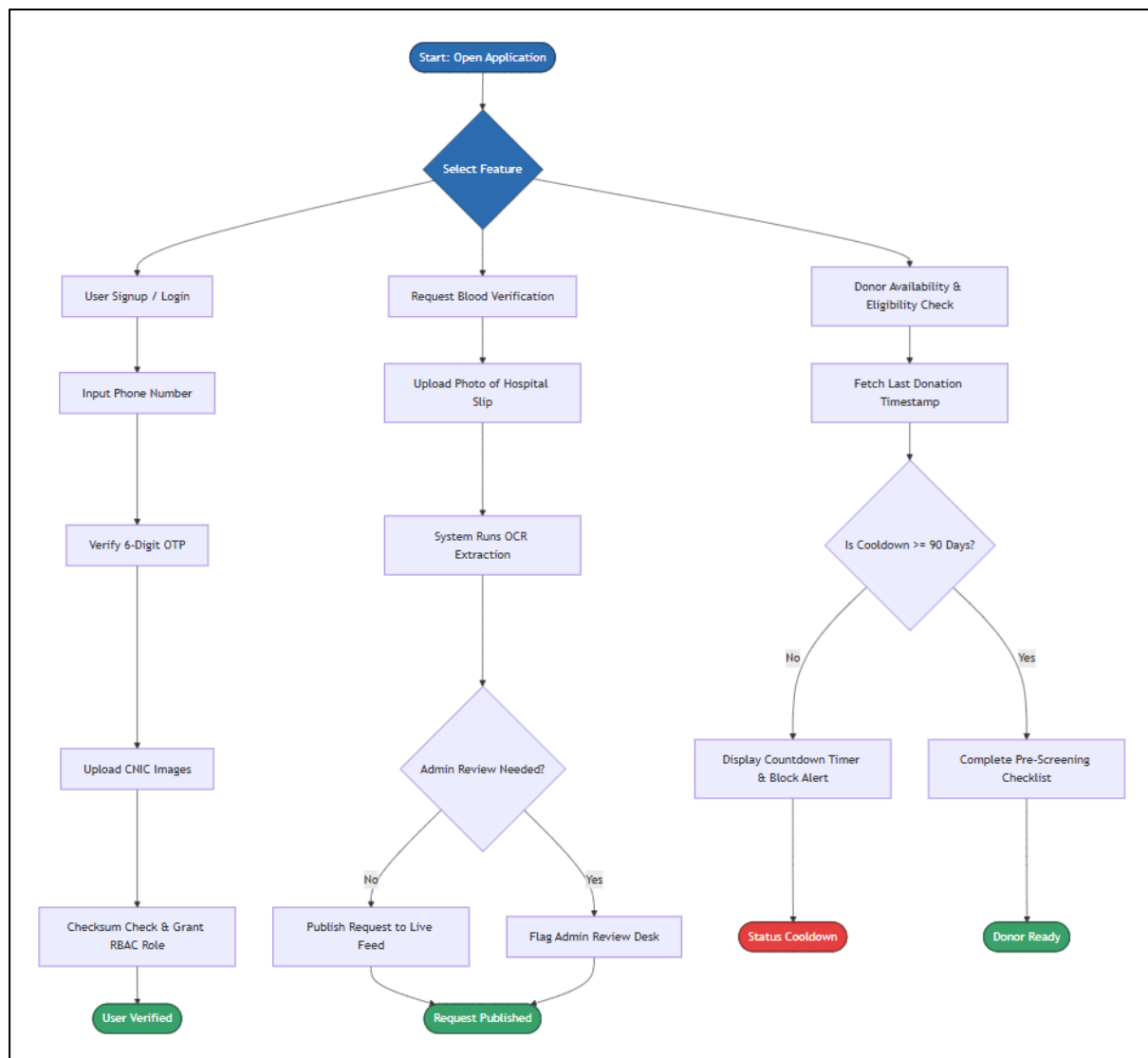
#### 4.4 Data Privacy, Security & Contact Masking

- Phone Number Masking (Proxy Routing): direct numbers are never exposed on the public feed or map; matched users communicate via masked proxy calling or end-to-end encrypted in-app messaging.
- CNIC Data Vault Encryption: captured CNIC documents and extracted numbers are stored in an isolated vault, AES-256 encrypted at rest, accessible only to automated verification services.

#### 4.5 Verification Pipeline Architecture



## 4.6 User Flow Diagram — Feature 2



## 4.7 Core System Use Case: Emergency Request Verification & Donor Matching

### 4.7.1 Use Case Specification

- Use Case ID: UC-SEC04-01
- Title: Validate Hospital Emergency Request & Dispatch Masked Donor Alert
- Primary Actor: Emergency Seeker
- Secondary Actors: Alkhidmat Admin, Verified Donor, OCR System Engine
- Preconditions: Seeker is authenticated via OTP and holds a valid registered CNIC account.
- Postconditions: Request is verified, cryptographically signed, published to the live map, and broadcast via proxy routing to eligible donors within 10 km.

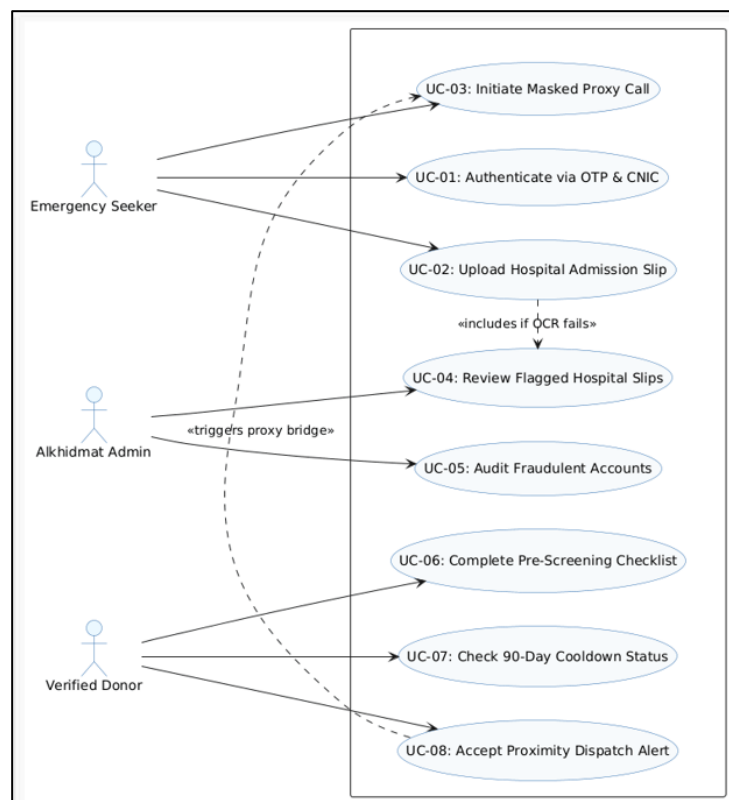
#### 4.7.2 Main Success Scenario

1. Seeker navigates to “Create Emergency Blood Request” and inputs blood group, hospital location, and unit requirements.
2. Seeker captures and uploads a clear photo of the hospital admission/requisition form.
3. OCR engine parses the document and extracts hospital name, doctor stamp, and MRN details.
4. System validates confidence score (>85%) and automatically approves the requisition.
5. System generates a verified emergency request card on the live feed.
6. System queries active donors within 10 km who have completed their 90-day cooldown and passed pre-screening.
7. Eligible donors receive an urgent push notification with hospital location and masked calling options.
8. A donor accepts; the system initiates proxy call bridging without exposing private phone numbers.

#### 4.7.3 Alternative & Exception Flows

- Alt Flow A (Low OCR confidence / blurred slip): request is flagged and routed to the 24/7 Verification Desk; an admin manually reviews against hospital databases within 3 minutes.
- Exception Flow B (Fraudulent / duplicate slip): if OCR detects a reused slip image or matching MRN within 24 hours, submission is auto-blocked, flagged, and the account is suspended pending inquiry.
- Exception Flow C (Donor failed pre-screening): if a donor reports recent fever/medication at the pre-screening popup, acceptance is politely restricted and an awareness appointment is suggested instead.

### 4.8 Use Case Diagram — Feature 2 Architecture



## 4.9 Module Summary & Key Performance Indicators

| Module Performance Metric      | Target KPI Standard             | System Impact                                                         |
|--------------------------------|---------------------------------|-----------------------------------------------------------------------|
| Average Verification Time      | < 3 minutes for 95% of requests | Ensures rapid emergency response without security bottlenecks.        |
| Fraudulent Request Rate        | 0% unverified posts on feed     | Restores public trust in digital blood broadcasts.                    |
| Donor On-Site Disqualification | Reduced by > 80%                | Pre-screening and cooldown engine filter out ineligible donors early. |
| Contact Privacy Breaches       | Zero direct phone exposures     | Protects female donors and the public from phone harassment.          |

## Section 5: Feature 3 — Social Media Feed for Blood Donation

### 5.1 Feature Overview

This feature is a dedicated in-app feed built only for blood donation. Today, most urgent requests spread through random WhatsApp forwards — messages get lost, copied incorrectly, or reach the wrong people too late. This feed replaces that chaos with one organized place where every post is structured, verified, and easy to act on. It has two parts: (1) Urgent Requests — real, verified requests, sorted and filterable, and (2) Awareness Content — drives, stories, and tips, so the app stays useful even outside of emergencies. Together it turns WhatsApp-style spam into a clean, trustworthy, one-stop feed.

### 5.2 Functional Requirements

#### FR 3.1 — Structured Post Creation

When someone needs blood, they fill a structured form rather than typing a free-text message. The form captures:

- Blood group needed
- Number of units needed
- Hospital name
- Urgency level (e.g. within 2 hours, within 24 hours)
- Hospital slip attachment (verified by the Section 4 pipeline before the post goes live)

#### FR 3.2 — Feed Filtering

Donors filter posts instead of scrolling everything, by:

- Blood group
- Urgency
- Location
- Drive events (upcoming blood donation drives)

#### FR 3.3 — Post Interactivity

- A one-tap “I Can Donate” button — the fastest possible response path.
- A share button so a post can be shared via direct link or WhatsApp without losing structure or getting garbled like a normal forward.

#### FR 3.4 — Request Lifecycle Management

Once enough donors have responded and required units are fulfilled, the post automatically closes, so donors don't show up to a hospital for a request that's already been met.

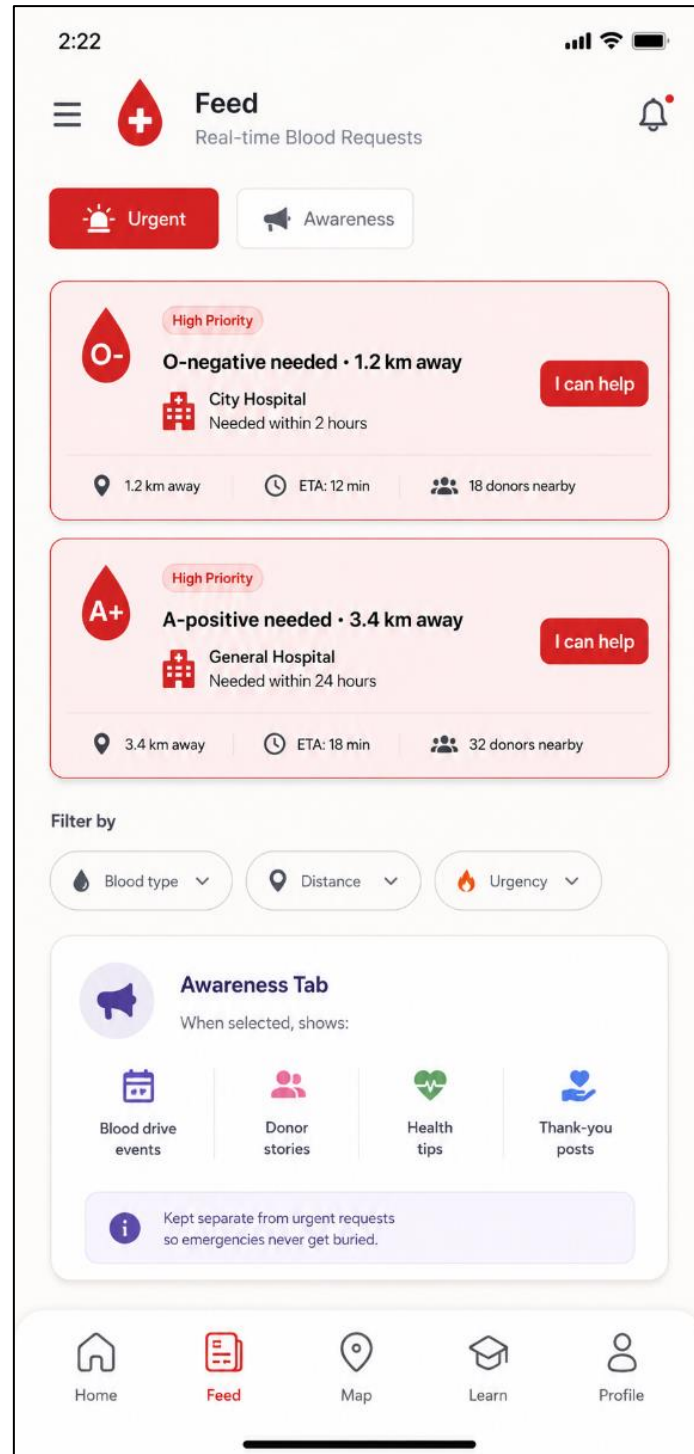
### 5.3 Notification Triggers

- Rare blood groups (e.g. O-negative, AB-negative): the system sends a push notification directly to all matching, eligible, nearby donors, since rare types are harder to find and need faster alerts.

- Common blood groups: notifications can be slightly less aggressive, since more donors are usually available.

## 5.4 Summary

Turn messy WhatsApp blood requests into a clean, verified feed where donors can filter what they see, respond in one tap, and stop getting notified once a request is already fulfilled — with extra-fast alerts for rare blood types.



## Section 6: Feature 4 — Awareness Sessions & Eligibility Module

### 6.1 Feature Overview

The Awareness Sessions & Eligibility Module improves public awareness about blood donation and helps users understand whether they're currently eligible to donate. It provides an in-app educational hub with reliable content, myth-busting material, and campus/community awareness sessions. It also includes an Interactive Eligibility Checker — a short four-step questionnaire that gives users a preliminary read on same-day eligibility before they commit to visiting a drive. Users can additionally browse upcoming drives and awareness sessions, register as donors or volunteers, and view basic health-screening feedback after donation.

#### Objectives

- Increase awareness about the importance of voluntary blood donation.
- Provide accessible educational material about the donation process.
- Address common myths and misconceptions.
- Provide a quick eligibility check before donation.
- Encourage participation in campus and community blood drives.
- Allow users to register as donors or volunteers for upcoming events.
- Provide basic health-screening feedback after donation.

### 6.2 Functional Requirements

#### FR 4.1 — Interactive Eligibility Checker

A quick four-step quiz to check same-day donation eligibility, intended as a pre-screening mechanism — not a medical determination — before a user attends a donation activity.

- Step 1 — Basic Eligibility Information: age and other core screening details.
- Step 2 — Recent Health Condition: recent illness or health condition that may affect eligibility.
- Step 3 — Recent Donation / Recovery Status: whether the user has donated recently or needs more recovery time.
- Step 4 — Final Eligibility Result: Eligible to Proceed / May Need Medical Confirmation / Not Eligible at This Time — with a clear disclaimer that final eligibility is set by the medical screening process on-site.

Expected benefits: fewer unnecessary drive visits, better donor understanding of basic eligibility, stronger pre-screening of the donor pool, and more efficient drive-day operations.

#### FR 4.2 — Awareness Content Library

Educational material addressing common misconceptions, organized into categories:

- Blood Donation Basics: importance, process, and blood-group information.
- Donation Myths and Facts: common misconceptions and evidence-based corrections, plus FAQs.

- Health and Donation Awareness: preparing for donation, following medical staff instructions, and post-donation care.
- Social and Cultural Awareness: addressing social, gender-related, and cultural misconceptions, and encouraging voluntary community-based donation.

Supported content types: articles, short educational videos, FAQs, myth-vs-fact content, and awareness session announcements. Users can browse, select a topic, read/watch content, and share it where appropriate.

### FR 4.3 — Event Registration & Booking

Users browse upcoming blood drives and awareness sessions and register as a donor or volunteer. Each event lists title, type, date/time, location, organizer, description, registration option, and available slots where applicable.

Registration flow: Browse Events → Select Event → View Event Details → Select Donor/Volunteer → Confirm Registration → Registration Confirmation. On success, the system stores the registration and shows a confirmation message.

Benefits: easier drive discovery, more participation in awareness activities, less dependence on unorganized communication channels, and organized participant records for organizers.

### FR 4.4 — Health Feedback & Reports

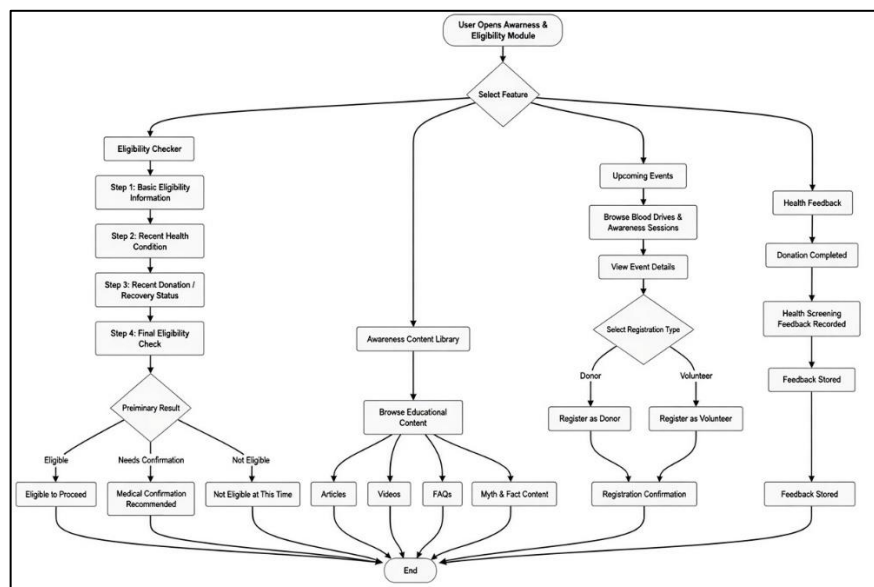
Stores and displays basic health-screening feedback after a donation — screening outcome, general feedback, post-donation instructions, follow-up recommendations, and donation date — limited to what's necessary for the feature, with appropriate protection of user information.

Flow: Donation Completed → Screening Feedback Recorded → Feedback Stored → User Views Feedback.

## 6.3 Awareness Sessions

Awareness sessions can be organized for universities and campuses, community groups, NGOs, student societies, volunteer groups, and other community organizations, covering: importance of blood donation, the donation process, donor eligibility, myths and facts, benefits of voluntary donation, blood group awareness, emergency blood requirements, how individuals can support emergency requests, and social/cultural barriers to donation. Event details are published through this module for discovery and registration.

## 6.4 User Flow Diagram





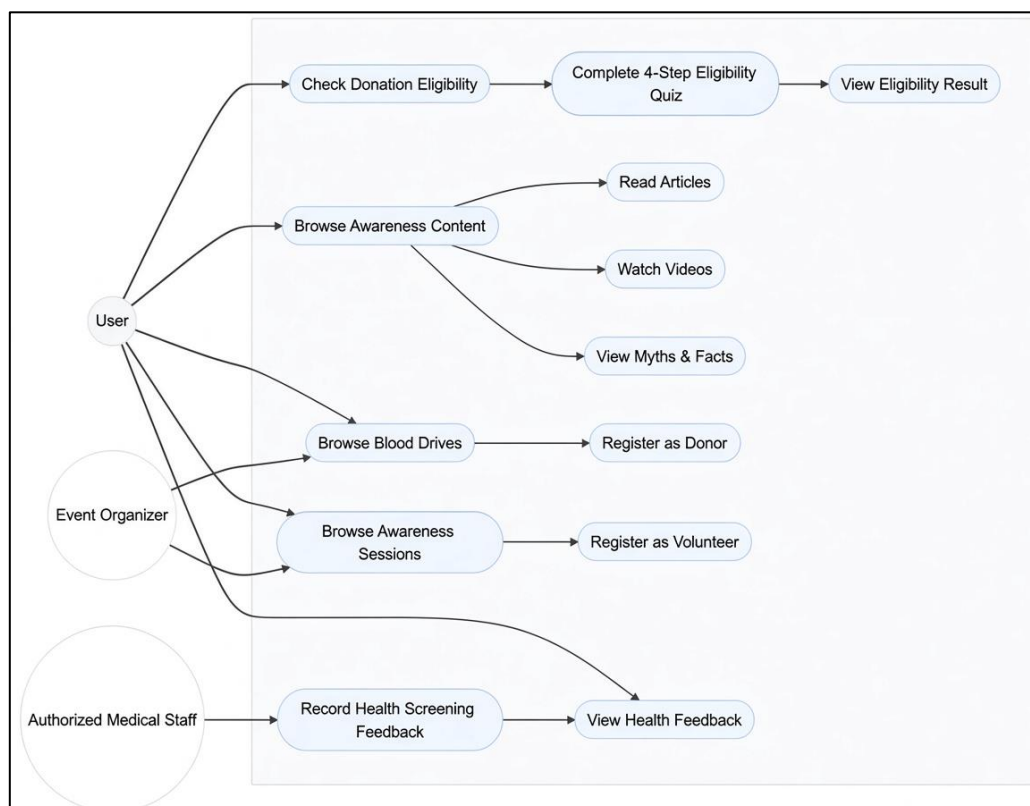
## 6.5 Module Components

| Component                 | Purpose                                      |
|---------------------------|----------------------------------------------|
| Eligibility Checker       | Quick preliminary donation eligibility check |
| Awareness Content Library | Educational blood donation content           |
| Myth & Fact Section       | Addresses common misconceptions              |
| Awareness Sessions        | Information about educational sessions       |
| Event Registration        | Register for blood drives and sessions       |
| Donor Registration        | Register as a potential donor                |
| Volunteer Registration    | Participate as a volunteer                   |
| Health Feedback           | Basic post-donation screening feedback       |

## 6.6 Example Use Case: User Checks Eligibility and Registers for a Blood Drive

Actor: Registered User.

Precondition: user has access to the Emergency Blood Response Platform.



Alternative flow: if the eligibility checker indicates the user may not currently be eligible, the system informs them not to proceed on the preliminary result alone and to follow appropriate medical guidance.

## **6.7 Expected Impact**

- Improved awareness about voluntary blood donation.
- Reduced misinformation and common misconceptions.
- Better user understanding of basic eligibility requirements.
- Stronger pre-screening ahead of donation activities.
- Increased participation in blood drives and awareness sessions.
- Greater community and campus involvement.
- Basic post-donation feedback made available to users.

## **6.8 Summary**

The Awareness Sessions & Eligibility Module combines education, preliminary eligibility screening, event participation, and post-donation feedback within a single feature. It supports the platform's wider goal of transforming blood donation from an emergency-only activity into a structured, continuously supported process.

## Section 7: Non-Functional Requirements & System KPIs

### 7.1 Section Overview

Sections 3 through 6 cover what the app does — the map, verification, the feed, awareness sessions. This section covers how well it needs to do all of that: the speed, security, and reliability standards that apply across the whole platform. This matters more here than in a typical app because people will be using this during real emergencies — a slow app when someone urgently needs blood, or a leaked CNIC or health record, is not just a bad review but a real problem for a real person.

#### Objectives

- Keep the platform usable even when traffic spikes during an emergency.
- Protect donor/seeker data — CNIC, health info, hospital slips — from exposure or misuse.
- Make sure access control is role-based and actually enforced, not just hidden in the UI.
- Set KPIs the team can check after launch, not just “did we build the feature.”
- Make sure the app doesn't fully break if one part (like the map) goes down.

### 7.2 Performance & Scalability Requirements

#### NFR 1.1 — Concurrent Users

Handle up to 10,000 concurrent users during peak periods — e.g. after a mass-casualty incident or when a rare-blood-group post starts spreading — without noticeable slowdown in search, posting, or matching. Achieved via horizontal scaling: multiple app server instances behind a load balancer, auto-scaling with traffic.

#### NFR 1.2 — Response Times

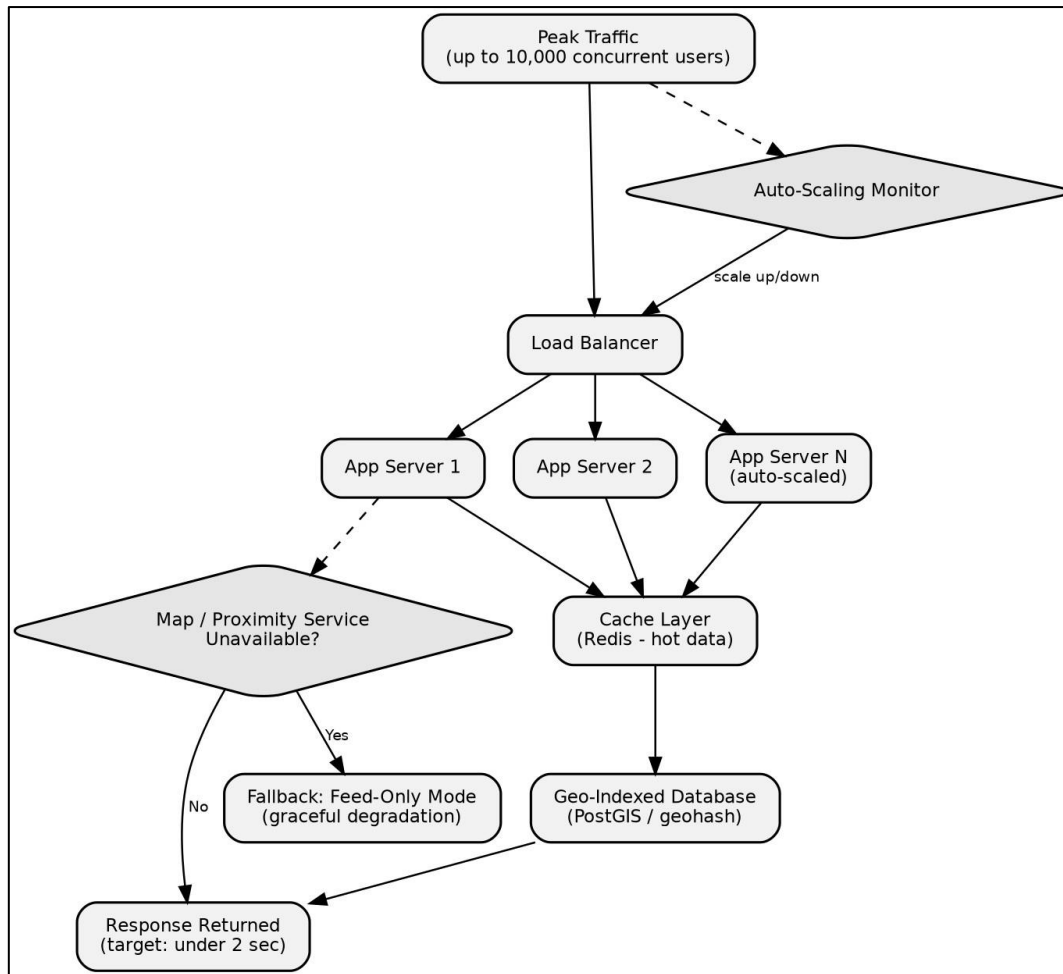
- Map and “find donors nearby” queries: under 2 seconds for at least 95% of requests.
- Posting a request, matching a donor, or verifying an OTP: roughly 3 seconds under normal conditions.

#### NFR 1.3 — Database & Caching

- Proximity search needs a geo-indexed database (e.g. PostGIS or geohashing) so nearest-donor lookups don't degrade as the donor pool grows into the thousands.
- A caching layer (e.g. Redis) keeps frequently requested data — like the live feed — from hitting the database on every read.

#### NFR 1.4 — Uptime & Fallback

- Target uptime of approximately 99.5%, excluding planned maintenance.
- If the map/proximity service goes down, the app should still allow posting and browsing requests through the feed rather than fully breaking.



### 7.3 Security & Data Protection Requirements

The platform handles sensitive material — CNIC numbers, health screening answers, hospital slips. A leak here is a much bigger deal than a typical data breach, so this section is treated as a first-class requirement rather than an afterthought.

#### NFR 2.1 — Encryption

- AES-256 for sensitive data at rest — CNIC, health data, hospital slip images.
- TLS 1.2+ (HTTPS) for everything moving between the app and the backend.

#### NFR 2.2 — Keeping Sensitive Data Separate

CNIC and health data sit in their own access-restricted store rather than being mixed with regular app tables, so a bug elsewhere in the app can't accidentally expose them.

#### NFR 2.3 — Contact Privacy (ties into Section 4)

Phone numbers stay masked at the API level — seekers and donors communicate through the app's masked/chat system until a request is actually accepted.

#### NFR 2.4 — Role-Based Access

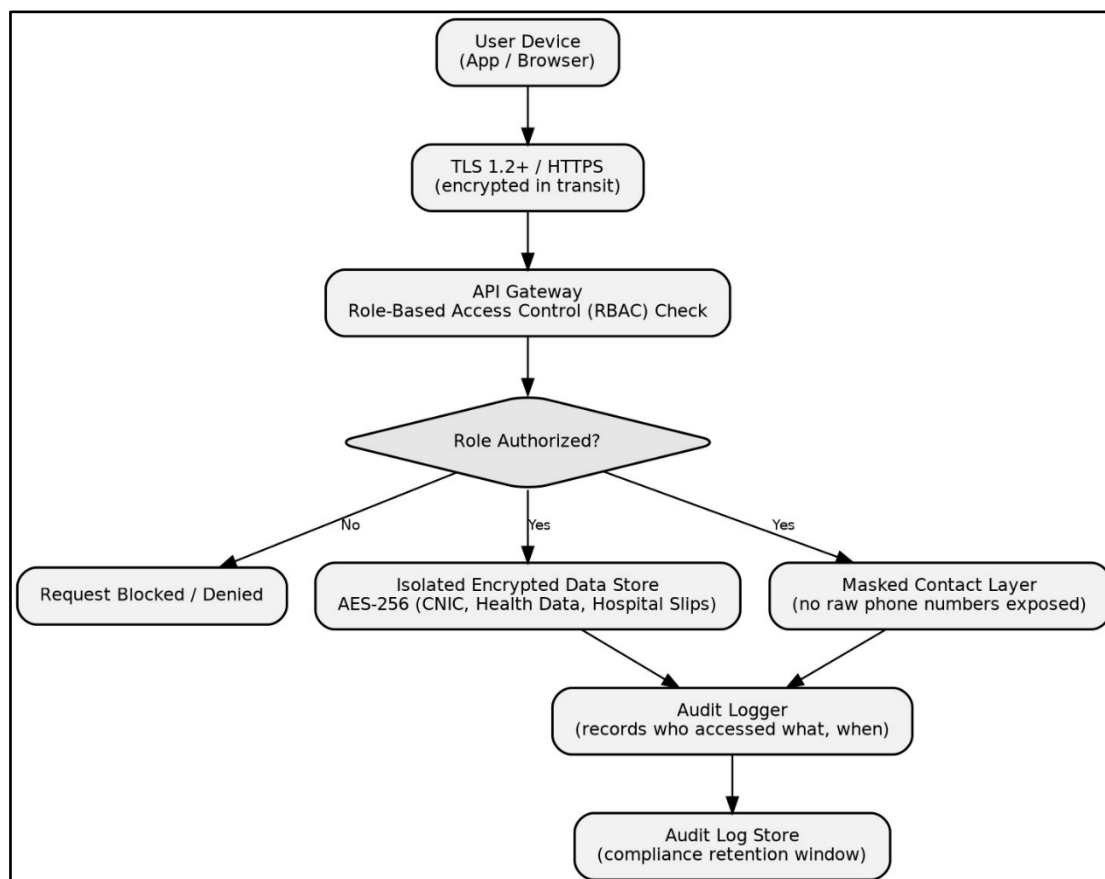
Admin, Verified Seeker, Verified Donor, and Drive Organizer permissions are enforced server-side, not just hidden via UI — otherwise someone could call the API directly and bypass the restriction.

## NFR 2.5 — Logging & Compliance

- Any access to sensitive records — CNIC checks, health data, admin actions — is logged with who did it and when, retained for a set period.
- Follows local medical/data privacy practices and Pakistan's data protection framework, including deleting data for accounts that go inactive.

## NFR 2.6 — File Handling & API Safety

- Uploaded hospital slips and CNIC images are scanned, size/type restricted, and stored encrypted — never on a publicly guessable URL.
- Rate limiting on OTP and login endpoints, input validation everywhere, and general OWASP Top 10 hygiene against injection-style attacks.
- A security review / penetration test before any major release, given how sensitive the data is.



## 7.4 Success Metrics (KPIs)

These are the numbers the team would actually check after the app is live to see if it's doing its job — not just whether the features exist.

| KPI                   | Target                           | Why It Matters                                                           |
|-----------------------|----------------------------------|--------------------------------------------------------------------------|
| Average Time-to-Match | Under 15 min for urgent requests | This is the whole point of the app — beating a WhatsApp broadcast chain. |

| KPI                        | Target                                                     | Why It Matters                                                                                  |
|----------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Verification Accuracy      | 0% unverified requests on the feed                         | If this fails, the Section 4 verification pipeline isn't actually working.                      |
| Donor Retention Rate       | % of donors who respond after their 90-day cooldown resets | Shows whether people keep coming back instead of donating once and disappearing.                |
| System Uptime              | $\geq 99.5\%$ monthly                                      | Tells us if the app is actually available when it's needed.                                     |
| Notification Delivery Rate | Geo-fenced alerts delivered within 60 sec                  | Checks that Section 3's alert system is actually reaching people in time.                       |
| Request Fulfillment Ratio  | % of requests fully met within 24 hrs                      | A bigger-picture number — not just fast first contact, but requests actually getting fulfilled. |
| Crash-Free Sessions        | $\geq 99\%$                                                | Basic stability check.                                                                          |

## 7.5 Non-Functional Components

These are the pieces sitting underneath Sections 3–6 that make the performance and security requirements above actually possible:

| Component             | What It's For                                                                     |
|-----------------------|-----------------------------------------------------------------------------------|
| Load Balancer         | Spreads incoming traffic across app servers so no single server gets overwhelmed. |
| Auto-Scaling Monitor  | Adds/removes server capacity based on current traffic.                            |
| Cache Layer (Redis)   | Speeds up frequently requested data, less load on the database.                   |
| Geo-Indexed Database  | Keeps location/proximity searches fast as the donor pool grows.                   |
| Encryption Layer      | AES-256 + TLS — protects data at rest and in transit.                             |
| RBAC Engine           | Enforces who can access what, server-side.                                        |
| Audit Logger          | Tracks access to sensitive data for accountability.                               |
| Monitoring & Alerting | Watches uptime/latency/errors and flags problems early.                           |

## 7.6 Example: What Happens Under a Traffic Spike

Scenario: a mass-casualty event happens and app usage spikes across the city within minutes.

- Traffic monitoring picks up the concurrent user count climbing toward the 10,000 mark.
- Auto-scaling kicks in and spins up extra app server instances behind the load balancer.
- Proximity search still returns in under 2 seconds because the database is geo-indexed instead of scanning every donor record one by one.
- The cache layer takes a lot of the repeated feed reads off the database.
- If the map API itself struggles or goes down under the load, the app switches to feed-only mode instead of just failing.
- Even during all this, CNIC/health record access still gets logged and still needs RBAC approval — that part doesn't get skipped just because traffic is high.

This is basically the whole reason this section exists — it's the difference between the app working when it's actually needed versus falling over at the worst possible time.

## 7.7 Expected Impact

- Keeps the app usable during the exact high-traffic moments it's built for.
- Protects donor/seeker trust by keeping CNIC and health data locked down.
- Gives the dev team actual numbers to build toward instead of vague goals like “make it fast.”
- Gives the team something concrete to report back to Alkhidmat supervisors after launch.
- Lowers the risk of the app breaking or losing user trust after release.

## 7.8 Summary

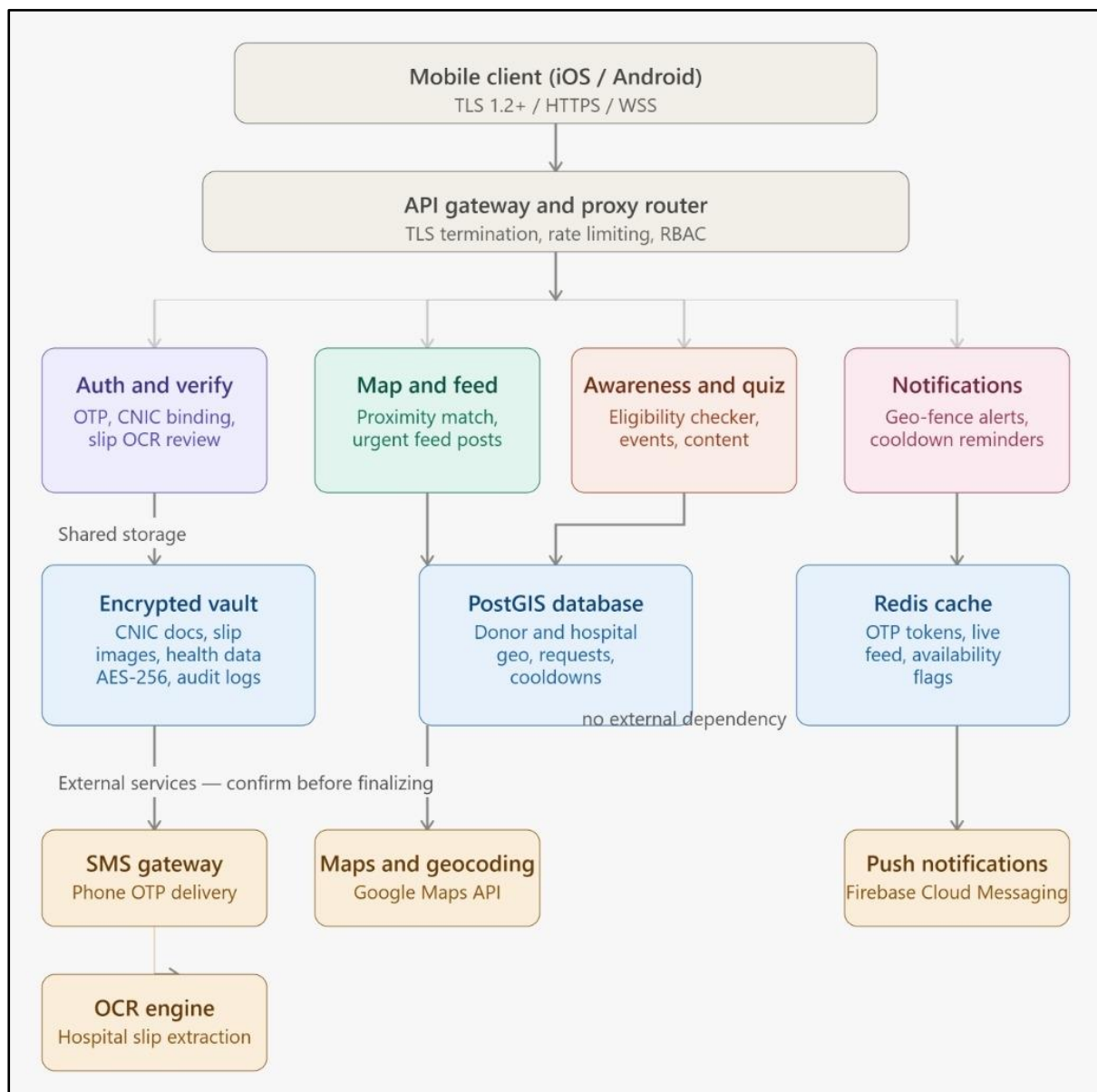
Section 7 sets the performance, security, and success metrics that apply across the whole platform. 7.2 covers keeping the app fast and available under emergency-level traffic. 7.3 covers protecting CNIC, health, and contact data through encryption, access control, and logging. 7.4 lays out the actual numbers — like a 15-minute time-to-match and 0% unverified requests — that tell us whether the app is working the way it's supposed to.

Basically, this section is what turns the features in Sections 3–6 from something that works in a demo into something that can actually be trusted during a real emergency.

## Section 8: System Architecture & Integration

### 8.1 High-Level Data Flow

The platform is organized around four feature modules — Live Map (Section 3), Authentication & Verification (Section 4), Social Feed (Section 5), and Awareness & Eligibility (Section 6) — sitting on a shared backend that owns identity, the geo-indexed donor/request store, and the notification pipeline. Verification (Section 4) is the gate every request passes through before it can reach the Map or the Feed; the cooldown and pre-screening state it maintains is read by both of those modules when deciding who to alert. The Awareness module feeds the same donor and event records that Verification and the Map rely on, so a donor who completes a campus drive registration or eligibility check is visible system-wide, not siloed to one module.





## 8.2 Consolidation & Document Sign-Off

This Project Requirement Document (PRD) provides the foundational blueprint for the development and deployment of the Alkhidmat Foundation Pakistan Emergency Blood Response Platform, Version 1.0. This document serves as the official integration of individual component specifications authored by the assigned owners of the Map, Verification, Feed, Awareness, and Non-Functional Requirement modules.

By signing off on this consolidated document, each Feature Owner validates that their specific module requirements are accurately represented, technically viable, and aligned with the cross-functional objectives of the system architecture described in Section 8.1. The KPI metrics identified in Section 7.4 establish the performance standards against which the success of Version 1.0 will be measured following its Karachi-specific launch.

Following this sign-off, this document becomes the primary reference for the platform's development, design, and QA engineering teams. Any subsequent architectural changes or functional adjustments will require a formal PRD amendment review to maintain system integrity.

| Role                 | Name                                  | Sign-Off                                                             | Date       |
|----------------------|---------------------------------------|----------------------------------------------------------------------|------------|
| Team Lead            | Abdul Hayy Khan                       | ✓ Approved                                                           | 11-08-2026 |
| Feature 1 Owner      | Hareem Israr                          | ✓ Approved                                                           | 11-08-2026 |
| Feature 2 Owner      | Saghir Ahmed                          | ✓ Approved                                                           | 11-08-2026 |
| Feature 3 Owner      | Mahrukh Baig                          | ✓ Approved                                                           | 11-08-2026 |
| Feature 4 Owner      | Yumna Abbasi                          | ✓ Approved                                                           | 11-08-2026 |
| NFR Owner            | Nimra Iftikhar                        | ✓ Approved                                                           | 11-08-2026 |
| Alkhidmat Supervisor | Sir Talha Shahid /<br>Sir Hamas Malik | <input type="checkbox"/> Pending<br><input type="checkbox"/> Pending |            |